

A SHIFTED INDEFINITE THETA COMPLETION OVER $\mathbb{Q}(\sqrt{5})$: AN ASYMMETRIC BOUNDARY DERIVATIVE AND A ROW-MODEL FACTORIZATION

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ABSTRACT. We study a norm-supported cone series $B(q)$ over $\mathbb{Q}(\sqrt{5})$ and two structures attached to it. First, a coefficientwise row model motivated by the even-row missing kernel in Chan's Θ_{10} dissection factors as $-q^{-1}j(q; q^5)^2B(q)$. Second, the normalized series $F(\tau) = q^{1/10}B(q)$ is the holomorphic sign kernel of an explicit rank-two indefinite theta datum in the sense of Zwegers. We write the exact characteristic vectors and the resulting componentwise weight-one transformation formulas; the five characteristic components in the S -image reduce algebraically to the two displayed components. One boundary correction vanishes by an odd-theta pairing. For the other boundary we give an index-four orthogonal decomposition and compute $\partial_{\bar{\tau}}\widehat{F}$ as a nonzero sum of a holomorphic weight- $\frac{1}{2}$ theta factor times an antiholomorphic unary weight- $\frac{3}{2}$ theta factor. The corresponding ξ_1 image is nonholomorphic, so this natural completion is not harmonic and does not by itself establish ordinary weight-one mock modularity in the harmonic-Maass sense. The result is instead an explicit shifted indefinite theta completion with a mixed theta derivative. The arithmetic theory of B is independent and belongs to a separate companion paper.

1. INTRODUCTION

The identities of Andrews, Dyson, and Hickerson and of Cohen connect q -series, real quadratic fields, and Maass waveforms [1, 6]. Their coefficients admit an ideal-theoretic interpretation and display the multiplicativity expected from a Hecke character. Related real-quadratic constructions were developed in [7, 3, 10]. The series studied here arises from a different source: Chan's dissection of the tenth-order theta function Θ_{10} [5].

The companion arithmetic paper [9] studies the independently defined cone series

$$B(q) = \sum_{N \geq 0} B_N q^N$$

and proves the finite lattice, norm, and coefficient identities needed to define it. This paper studies the analytic nature of that already defined series. Its main point is that the natural completion has one vanishing boundary correction and one surviving boundary correction. The asymmetry is visible both in the unequal norms -5 and -20 of the two negative vectors and in the parity of the transverse theta sums.

Our principal analytic result is the following. The theta components θ_j and g_j are defined explicitly in Section 7.

Theorem 1.1 (Exact completion and differential image). *Let $F(\tau) := q^{1/10}B(q)$, where $q = e^{2\pi i\tau}$. For*

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -5 \end{pmatrix}, \quad a = \left(\frac{1}{2}, \frac{1}{10}\right), \quad b = \left(\frac{1}{2}, -\frac{1}{10}\right),$$

and the negative-vector pair $c_2 = (-5, 3)$, $c_1 = (0, 1)$, let $\vartheta_{a,b}^{c_2, c_1}$ denote Zwegers' completed indefinite theta function for $Q_0(v) = \frac{1}{2}v^tAv$. Then

$$\widehat{F}(\tau) := \frac{1}{2}e^{-3\pi i/5}\vartheta_{a,b}^{c_2, c_1}(\tau)$$

has holomorphic sign part F and satisfies

$$\widehat{F}(\tau + 1) = e^{\pi i/5}\widehat{F}(\tau),$$

$$\widehat{F}(-1/\tau) = \frac{\tau}{2\sqrt{5}} \sum_{m=0}^4 \vartheta_{b+(0, m/5), -a}^{c_2, c_1}(\tau).$$

Writing $a'_m = b + (0, m/5)$, this five-term expression reduces to

$$\widehat{F}(-1/\tau) = \frac{\tau}{2\sqrt{5}} \left[(1 + e^{-\pi i/5})\vartheta_{a'_1, -a}^{c_2, c_1}(\tau) + (1 + e^{3\pi i/5})\vartheta_{a'_4, -a}^{c_2, c_1}(\tau) \right].$$

Writing $Y = \text{Im}(\tau)$, its differential images are

$$\frac{\partial \widehat{F}}{\partial \bar{\tau}} = -\frac{i}{4\sqrt{10Y}} \sum_{j=0}^3 (-1)^j \theta_j(\tau) \overline{g_j(\tau)},$$

and, for $\xi_1(f) = 2iY \overline{\partial f / \partial \bar{\tau}}$,

$$\xi_1(\widehat{F}) = -\frac{\sqrt{Y}}{2\sqrt{10}} \sum_{j=0}^3 (-1)^j \overline{\theta_j(\tau)} g_j(\tau).$$

These expressions are nonzero and the ξ_1 image is not holomorphic. Hence $\widehat{F}$ is not a weight-one harmonic Maass form. The displayed S - and T -identities do not by themselves specify a closed vector-valued representation.

Here

$$j(z; q) := (z; q)_\infty (q/z; q)_\infty (q; q)_\infty$$

is Jacobi's product. We do not attach a stronger mock-modular label to the residual product here: doing so requires carrying the exact monomial normalization and the vector-valued representation through the Jacobi factor. The formal product identity is stated independently in Theorem 5.7.

Relation to earlier work. Zwegers' thesis supplies the general indefinite-theta completion and its characteristic transformation formulas [12]. His later paper on mock Maass theta functions places ADH- and Cohen-type sums in a broader real-analytic setting [11]. The contribution here is the exact specialization for this shifted $\mathbb{Q}(\sqrt{5})$ cone: the characteristics, the asymmetric boundary decomposition, and the mixed theta derivative. Hickerson–Mortenson's treatment of Hecke-type double sums gives complementary q -series background [8]. We do not infer a harmonic-Maass or ordinary mock classification merely from the existence of a Zwegers completion.

Organization. Section 2 recalls only the definitions needed from the companion paper. Section 3 gives a brief arithmetic separation statement. Section 4 proves the slab identity, Section 6 gives the corrected Pell coordinates and sign calculation, and Section 7 computes the completion and mixed boundary derivative. Section 8 compares the construction with the Maass cases, and Section 9 records computation, formalization scope, and open questions.

2. THE RESIDUAL SERIES RECALLED FROM THE COMPANION PAPER

We keep the common foundation short and refer to [9] for the lattice bijection, norm interpretation, unit action, prime classification, and their proofs.

Definition 2.1. For $k, r \in \mathbb{Z}$, put

$$H(k, r) := 4k^2 + 2k + r^2 + (6k + 1)r, \quad E(k, r) := \frac{H(k, r)}{2}.$$

The integer $H(k, r)$ is even. Define the two quadrant cones

$$\mathcal{A} := \{(k, r) \in \mathbb{Z}^2 : k \geq 0, r \geq 0\}, \quad \mathcal{D} := \{(k, r) \in \mathbb{Z}^2 : k < 0, r < 0\}.$$

On these two cones $E(k, r) \geq 0$.

Definition 2.2. The coefficient and generating series are

$$B_N := \sum_{\substack{(k, r) \in \mathcal{D} \\ E(k, r) = N}} (-1)^r - \sum_{\substack{(k, r) \in \mathcal{A} \\ E(k, r) = N}} (-1)^r, \quad B(q) := \sum_{N \geq 0} B_N q^N.$$

Each coefficient is a finite signed sum.

The companion paper proves that the map

$$\beta(k, r) = (r - 2k) + (4k + 3r + 1)\varphi$$

is a bijection from $\mathbb{Z}^2$ to a translate of an index-10 sublattice of $\mathbb{Z}[\varphi]$ and that

$$-N(\beta(k, r)) = 10E(k, r) + 1.$$

Section 5 defines a coefficientwise row model motivated by an even-row missing-kernel expression in the Θ_{10} project and proves that the model factors exactly as

$$\mathcal{R}(q) = -q^{-1}j(q; q^5)^2 B(q)$$

in the E -level exponent chart (Theorem 5.7; an earlier project record displayed this identity with a mixed normalization — see Remark 5.8). The current formal theorem is about the row model defined here; an identification with the older source object `rhoEvenSourceOldRowMissing` would require a separate source-to-model theorem and is not claimed. The finite algebraic results above are inputs here, not reproved in this paper.

Remark 2.3 (Hickerson–Mortenson form). In the notation of [8], B coincides with the Hecke-type double sum

$$-f_{1,3,4}(X, -X^3, X);$$

this follows exactly by termwise comparison of the two cones, signs, and exponents. The coefficient implementation was additionally checked through $N = 200$.

3. ARITHMETIC SEPARATION IN FIVE SENTENCES

The companion paper identifies the defining translate in $\mathbb{Z}[\varphi]$ and proves the norm formula above [9]. The fundamental totally positive unit does not stabilize that translate at one step, so the element sum does not descend by the classical ADH route to a single ideal character. The resulting coefficient sequence is nonmultiplicative, with $B_{34} = 3$ but $B_1 B_3 = -2$. This witness rules out the classical one-dimensional Hecke-character explanation; it does not rule out every automorphic interpretation. The prime-value classification and all sector-residue questions belong to the companion paper and are not duplicated here [9].

4. THE INVOLUTION AND THE EXACT SLAB IDENTITY

The analytic form of B becomes visible after a sign-reversing involution removes the paired interior.

Definition 4.1. For fixed $k \in \mathbb{Z}$, define

$$\sigma_k(r) := -r - 6k - 1.$$

Define the two slabs

$$\begin{aligned} \mathcal{S}^+ &:= \{(k, r) \in \mathbb{Z}^2 : k \geq 0, r \leq -6k - 1\}, \\ \mathcal{S}^- &:= \{(k, r) \in \mathbb{Z}^2 : k \leq -1, r \geq -6k\}. \end{aligned}$$

Lemma 4.2 (The involution). *For every $k, r \in \mathbb{Z}$,*

$$\sigma_k(\sigma_k(r)) = r, \quad E(k, \sigma_k(r)) = E(k, r), \quad (-1)^{\sigma_k(r)} = -(-1)^r.$$

The map σ_k has no fixed point in $\mathbb{Z}$.

Proof. Write $C = 6k + 1$. Since

$$H(k, r) = 2k(2k + 1) + r(r + C),$$

we have

$$(-C - r)(-r) = r(r + C),$$

which proves preservation of H and hence of E . The number C is odd, so $(-1)^{-C-r} = -(-1)^r$. A fixed point would give $2r = -C$, impossible because the right side is odd. $\square$

Theorem 4.3 (Slab identity). *Coefficientwise as a formal power series,*

$$B(q) = \sum_{(k,r) \in \mathcal{S}^+} (-1)^r q^{E(k,r)} - \sum_{(k,r) \in \mathcal{S}^-} (-1)^r q^{E(k,r)}.$$

Proof. For $(k, r) \in \mathcal{A}$, the point $(k, \sigma_k(r))$ lies in $\mathcal{S}^+$, and this assignment is a bijection from $\mathcal{A}$ to $\mathcal{S}^+$. Lemma 4.2 gives

$$-(-1)^r q^{E(k,r)} = (-1)^{\sigma_k(r)} q^{E(k, \sigma_k(r))}.$$

For $(k, r) \in \mathcal{D}$, the point $(k, \sigma_k(r))$ lies in $\mathcal{S}^-$, and the same map is a bijection from $\mathcal{D}$ to $\mathcal{S}^-$. In this case

$$(+1)(-1)^r q^{E(k,r)} = -(-1)^{\sigma_k(r)} q^{E(k, \sigma_k(r))}.$$

Adding the two identities proves the result coefficientwise. No analytic rearrangement is used. $\square$

5. A ROW-MODEL KERNEL AND ITS EXACT FACTORIZATION

Throughout, all series are formal Laurent series over $\mathbb{Z}$: a series is *admissible* if its exponents are bounded below and each coefficient is a finite integer sum. Recall from Definition 2.1 the quadratics

$$H(k, r) = 4k^2 + 2k + r^2 + (6k + 1)r, \quad E(k, r) = \frac{1}{2}H(k, r),$$

the cones $\mathcal{A} = \{k \geq 0, r \geq 0\}$, $\mathcal{D} = \{k < 0, r < 0\}$, the coefficients B_N of Definition 2.2, and the involution $\sigma_k(r) = -r - 6k - 1$ of Lemma 4.2. Set further

$$Q_{\text{out}}(u, v) := 5u^2 - 3u + 5v^2 - 7v.$$

Both Q_{out} and H take only even values, and $Q_{\text{out}}(u, v) \geq -2$ for all $u, v \in \mathbb{Z}$.

5.1. The row-model kernel. Motivated by the even-row expression in the Θ_{10} formalization, consider the four-fold row model with weight $(-1)^{u+v+r}$, exponent $Q_{\text{out}}(u, v) + H(k, r)$ (in the dissection chart the exponent carries an overall factor 9, which we normalize away), and (k, r) restricted to the two mixed quadrants, with a relative sign:

$$R_k := \begin{cases} \{r \leq -1\}, & k \geq 0, \\ \{r \geq 0\}, & k \leq -1, \end{cases} \quad \varepsilon_k := \begin{cases} +1, & k \geq 0, \\ -1, & k \leq -1. \end{cases}$$

The four-fold sum must be grouped correctly to be meaningful. For fixed H -value m the mixed quadrants may contain *infinitely many* lattice points: the value $m = -6$, for instance, is attained on $\{k \geq 0, r \leq -1\}$ at

$$(1, -3), (1, -4), (3, -3), (3, -16), (31, -24), (31, -163), \dots,$$

an infinite Pell-automorph orbit family; moreover $H(k, -3k) = -5k^2 - k \rightarrow -\infty$, so H is unbounded below there. Thus the naive sum over $\mathbb{Z}_{u,v}^2 \times (\text{mixed quadrants})$ is not absolutely summable coefficientwise. The computational probe `even_missing_kernel_term` and the Lean declaration `evenMissingKernelTerm` use the same row convention; the older Lean source object is `rhoEvenSourceOldRow_missing`. These definitions sum *row by row*: for each fixed k the inner variable r is resolved exactly. We adopt the same convention.

Definition 5.1 (Row-model kernel, row convention). For $k \in \mathbb{Z}$ define the row series

$$\rho_k(x) := \varepsilon_k \sum_{u \in \mathbb{Z}} \sum_{v \in \mathbb{Z}} \sum_{r \in R_k} (-1)^{u+v+r} x^{Q_{\text{out}}(u,v) + H(k,r)},$$

and the row-model kernel

$$\text{MK}(x) := \sum_{k \in \mathbb{Z}} \rho_k(x).$$

Lemma 5.4 below shows each ρ_k is admissible and the k -sum is locally finite, so MK is a well-defined Laurent series with integer coefficients. All its exponents are even.

5.2. Row transport. The engine of the proof is a row-local transport identity: on each row, the mixed-quadrant sum equals *minus* the corresponding cone sum. It refines the slab identity (Theorem 4.3) and is proved by the same involution.

Lemma 5.2 (Row transport). *Fix $k \in \mathbb{Z}$ and $m \in \mathbb{Z}$. All four sums below have finitely many terms, and*

$$\begin{aligned} \sum_{\substack{r \leq -1 \\ H(k,r)=m}} (-1)^r &= - \sum_{\substack{r \geq 0 \\ H(k,r)=m}} (-1)^r & (k \geq 0), \\ \sum_{\substack{r \geq 0 \\ H(k,r)=m}} (-1)^r &= - \sum_{\substack{r \leq -1 \\ H(k,r)=m}} (-1)^r & (k \leq -1). \end{aligned}$$

Equivalently, as admissible Laurent series in x ,

$$(1) \quad \sum_{r \in R_k} (-1)^r x^{H(k,r)} = - \sum_{r \in R_k^c} (-1)^r x^{H(k,r)},$$

where $R_k^c = \mathbb{Z} \setminus R_k$ is the cone side of the row ($r \geq 0$ if $k \geq 0$; $r \leq -1$ if $k \leq -1$).

Proof. Finiteness: for fixed k , $H(k, \cdot)$ is a monic quadratic in r , so each value is attained at most twice and $H(k, r) \rightarrow \infty$ as $|r| \rightarrow \infty$; in particular each row series is admissible.

Let $k \geq 0$ and recall $\sigma_k(r) = -r - 6k - 1$, which preserves $H(k, \cdot)$ and flips $(-1)^r$ (Lemma 4.2). Split

$$\{r \leq -1\} = S \sqcup T, \quad S := \{-6k \leq r \leq -1\}, \quad T := \{r \leq -6k - 1\}.$$

The strip S (empty when $k = 0$) is σ_k -stable: $r \in S$ gives $\sigma_k(r) = -r - 6k - 1 \in S$. Since σ_k has no fixed points, it partitions the finite set S into pairs with equal H -value and opposite sign, so S contributes 0 to every coefficient. Next, $r \geq 0 \iff \sigma_k(r) \leq -6k - 1$, so σ_k is a bijection $\{r \geq 0\} \rightarrow T$ preserving H and flipping the sign; hence the T -sum equals minus the $\{r \geq 0\}$ -sum, coefficient by coefficient.

For $k \leq -1$ the argument is the mirror image with $S = \{0 \leq r \leq -6k - 1\}$ (a σ_k -stable strip of even cardinality $-6k$) and $T = \{r \geq -6k\} = \sigma_k(\{r \leq -1\})$. $\square$

Remark 5.3. Summing (1) over $k \geq 0$ and $k \leq -1$ separately recovers the σ -straddling description of the row model and, in the E -variable, the slab identity of Theorem 4.3. The row-local form is what is needed here, because only after transport do the row supports become bounded below uniformly enough to sum over k .

5.3. Well-definedness and separation.

Lemma 5.4 (Well-definedness). *Each ρ_k is admissible, and*

$$(2) \quad \rho_k(x) = \begin{cases} -\Theta_u(x) \Theta_v(x) \sum_{r \geq 0} (-1)^r x^{H(k,r)}, & k \geq 0, \\ \Theta_u(x) \Theta_v(x) \sum_{r \leq -1} (-1)^r x^{H(k,r)}, & k \leq -1, \end{cases}$$

where

$$\Theta_u(x) := \sum_{u \in \mathbb{Z}} (-1)^u x^{5u^2 - 3u}, \quad \Theta_v(x) := \sum_{v \in \mathbb{Z}} (-1)^v x^{5v^2 - 7v}.$$

The support of ρ_k is contained in $[4k^2 + 2k - 2, \infty)$ for $k \geq 0$ and in $[4k^2 - 4k - 2, \infty)$ for $k \leq -1$. Consequently, for each exponent n only the rows with $4k^2 - 4|k| - 2 \leq n$ contribute, and $\text{MK} = \sum_k \rho_k$ is a well-defined Laurent series.

Proof. Fix k and an exponent n . A term of ρ_k contributes to x^n only if $Q_{\text{out}}(u, v) = a$ and $H(k, r) = m$ with $a + m = n$. Since $a \geq -2$ and $m \geq \min_{r \in R_k} H(k, r) > -\infty$, only finitely many splittings (a, m) occur; each a is attained by finitely many (u, v) (the quadratic Q_{out} is positive definite up to the bounded linear part: $5u^2 - 3u \leq a + 2$ and $5v^2 - 7v \leq a + 2$ bound u, v), and each m by at most two $r \in R_k$. Hence ρ_k is admissible, and by the distributive law for admissible series,

$$\rho_k(x) = \varepsilon_k \left(\sum_u (-1)^u x^{5u^2-3u} \right) \left(\sum_v (-1)^v x^{5v^2-7v} \right) \left(\sum_{r \in R_k} (-1)^r x^{H(k,r)} \right).$$

Here the index set is the full product $\mathbb{Z}_u \times \mathbb{Z}_v \times R_k$: the region constraint involves (k, r) only, the exponent is additively separated, and the weight $(-1)^{u+v+r}$ is multiplicatively separated — this is the entire content of the separation step. Applying Lemma 5.2 to the last factor gives (2).

Supports: for $k \geq 0$ the map $r \mapsto H(k, r)$ is increasing on $r \geq 0$, so $\min_{r \geq 0} H(k, r) = H(k, 0) = 4k^2 + 2k$; for $k \leq -1$ the vertex $-(6k + 1)/2$ of $H(k, \cdot)$ is positive, so on $r \leq -1$ the minimum is $H(k, -1) = 4k^2 - 4k$. Since $\Theta_u \Theta_v$ has exponents ≥ -2 , the stated supports follow from (2), and local finiteness of $\sum_k \rho_k$ is immediate. $\square$

Proposition 5.5 (Separation). *As Laurent series,*

$$\text{MK}(x) = \Theta_u(x) \Theta_v(x) (D(x) - A(x)) = \Theta_u(x) \Theta_v(x) B(x^2),$$

where $A(x) = \sum_{(k,r) \in \mathcal{A}} (-1)^r x^{H(k,r)}$ and $D(x) = \sum_{(k,r) \in \mathcal{D}} (-1)^r x^{H(k,r)}$.

Proof. Summing (2) over k : the sums $\sum_{k \geq 0} \sum_{r \geq 0}$ and $\sum_{k \leq -1} \sum_{r \leq -1}$ are the cone series $A(x)$ and $D(x)$, which are admissible with locally finite row decompositions (row supports $[4k^2 + 2k, \infty)$ resp. $[4k^2 - 4k, \infty)$ as in Lemma 5.4), so multiplication by the admissible series $\Theta_u \Theta_v$ commutes with the k -summation coefficientwise:

$$\text{MK} = \Theta_u \Theta_v \left(\sum_{k \leq -1} \sum_{r \leq -1} (-1)^r x^H - \sum_{k \geq 0} \sum_{r \geq 0} (-1)^r x^H \right) = \Theta_u \Theta_v (D - A).$$

Finally $H(k, r) = 2E(k, r)$, so $A(x) = \sum_N A_N x^{2N}$ and $D(x) = \sum_N D_N x^{2N}$ with A_N, D_N the cone counts of Definition 2.2; hence $D(x) - A(x) = B(x^2)$. $\square$

5.4. The theta legs.

Lemma 5.6 (Jacobi legs). *With $j(z; q) := \sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2}} z^n = (z; q)_\infty (q/z; q)_\infty (q; q)_\infty$,*

- (1) $\Theta_v(x) = -x^{-2} \Theta_u(x)$;
- (2) $\Theta_u(x) = j(x^2; x^{10}) = (x^2; x^{10})_\infty (x^8; x^{10})_\infty (x^{10}; x^{10})_\infty$;
- (3) $\Theta_u(x) \Theta_v(x) = -x^{-2} j(x^2; x^{10})^2$.

Proof. (i) Substitute $u = 1 - v$ in Θ_u : since $5(1 - v)^2 - 3(1 - v) = 5v^2 - 7v + 2$ and $(-1)^{1-v} = -(-1)^v$, the bilateral sum gives $\Theta_u(x) = -x^2 \Theta_v(x)$. (ii) $5u^2 - 3u = 10\binom{u}{2} + 2u$, so $\Theta_u(x) = \sum_u (-1)^u (x^{10})^{\binom{u}{2}} (x^2)^u = j(x^2; x^{10})$, and the Jacobi triple product evaluates it as stated. (iii) Multiply (i) and (ii). $\square$

5.5. Main theorem.

Theorem 5.7 (Row-model factorization). *As formal Laurent series,*

$$\text{MK}(x) = -x^{-2} j(x^2; x^{10})^2 B(x^2).$$

Equivalently, since every exponent of MK is even, writing $\mathcal{R}(q) := \sum_N [x^{2N}] \text{MK}(x) q^N$ for the row model in the halved (that is, E -level) variable,

$$\boxed{\mathcal{R}(q) = -q^{-1} j(q; q^5)^2 B(q)}$$

with $j(q; q^5) = (q; q^5)_\infty (q^4; q^5)_\infty (q^5; q^5)_\infty$.

Proof. Combine Proposition 5.5 with Lemma 5.6(iii); then substitute $q = x^2$, using $j(x^2; x^{10}) = j(q; q^5)|_{q=x^2}$. $\square$

Remark 5.8 (Normalization of the previously displayed identity). Earlier project records displayed the candidate identity as $\mathcal{R}(q) = -q^{-1} j(q; q^5)^2 B(q^2)$. That display mixes the two exponent charts: the theta legs are written at E -level (base q^5) while the cone factor is left at H -level. The internally consistent forms are the two displayed in Theorem 5.7; the hybrid form fails already at the constant term (-3 versus -2). The coefficientwise verification (now to $N = 2048$) confirms the corrected forms.

Remark 5.9 (Chart with the factor 9). In the dissection chart of the formal development the exponent is $9(Q_{\text{out}} + H) = 18 E_{\text{tot}}$. Thus the coefficient of q^{18N} there equals the coefficient of q^N in $-q^{-1} j(q; q^5)^2 B(q)$, and every exponent not divisible by 18 has coefficient 0 — in particular the kernel is supported on multiples of 9, consistent with the $9 \mid e$ phenomenon observed for the per-fiber counterexamples.

6. PELL COORDINATES AND THE ZWEGERS CONE

Set

$$x := r + 3k, \quad y := k, \quad P(X, Y) := X^2 - 5Y^2.$$

Thus $r = x - 3y$ and $k = y$.

Proposition 6.1 (Corrected shifted Pell identity). *With*

$$\mu := \left(\frac{1}{2}, \frac{1}{10} \right), \quad v_{k,r} := (x, y) + \mu,$$

we have

$$H(k, r) = x^2 + x - 5y^2 - y = P(v_{k,r}) - \frac{1}{5}$$

and hence

$$E(k, r) = \frac{1}{2} P(v_{k,r}) - \frac{1}{10}.$$

Moreover $(-1)^r = (-1)^{x+y}$.

Proof. Substituting $r = x - 3y$ and $k = y$ gives

$$4k^2 + 6kr + r^2 = x^2 - 5y^2, \quad 2k + r = x - y.$$

Therefore $H = x^2 + x - 5y^2 - y$. On the other hand,

$$P\left(x + \frac{1}{2}, y + \frac{1}{10}\right) = x^2 + x - 5y^2 - y + \frac{1}{5}.$$

The parity statement follows from $r = x - 3y \equiv x + y \pmod{2}$. $\square$

Let the bilinear form associated with P be

$$\mathcal{B}((X, Y), (X', Y')) := 2(XX' - 5YY').$$

Choose the two negative vectors

$$c_1 = (0, 1), \quad c_2 = (-5, 3),$$

for which

$$P(c_1) = -5, \quad P(c_2) = -20.$$

Proposition 6.2 (Corrected bilinear signs). *For $v_{k,r} = (x + \frac{1}{2}, y + \frac{1}{10})$,*

$$\mathcal{B}(v_{k,r}, c_1) = -(10k + 1)$$

and

$$\mathcal{B}(v_{k,r}, c_2) = -10(6k + r) - 8.$$

Neither expression vanishes for integral k, r .

Proof. Since $y = k$,

$$\mathcal{B}(v_{k,r}, c_1) = 2\left(0 - 5\left(k + \frac{1}{10}\right)\right) = -(10k + 1).$$

Since $x = r + 3k$ and $y = k$,

$$\begin{aligned} \mathcal{B}(v_{k,r}, c_2) &= 2\left[-5\left(x + \frac{1}{2}\right) - 15\left(y + \frac{1}{10}\right)\right] \\ &= -10x - 30y - 8 \\ &= -10(r + 6k) - 8. \end{aligned}$$

The first expression is odd, and the second is congruent to 2 modulo 10, so neither is zero. $\square$

Theorem 6.3 (The slab as a sign-difference cone). *For every $(k, r) \in \mathbb{Z}^2$,*

$$\mathbf{1}_{\mathcal{S}^+}(k, r) - \mathbf{1}_{\mathcal{S}^-}(k, r) = \frac{1}{2}(\operatorname{sgn} \mathcal{B}(v_{k,r}, c_2) - \operatorname{sgn} \mathcal{B}(v_{k,r}, c_1)).$$

Proof. By Proposition 6.2,

$$\operatorname{sgn} \mathcal{B}(v_{k,r}, c_1) = \begin{cases} -1, & k \geq 0, \\ +1, & k \leq -1, \end{cases}$$

and

$$\operatorname{sgn} \mathcal{B}(v_{k,r}, c_2) = \begin{cases} +1, & 6k + r \leq -1, \\ -1, & 6k + r \geq 0. \end{cases}$$

The four possible sign pairs give respectively the values 1, -1, and 0 off the two slabs, exactly as stated. $\square$

Put $q = e^{2\pi i\tau}$ with τ in the upper half-plane. Combining Theorem 4.3, Proposition 6.1, and Theorem 6.3 gives the exact expression

$$(3) \quad B(q) = q^{-1/10} \sum_{(x,y) \in \mathbb{Z}^2} (-1)^{x+y} \frac{\operatorname{sgn} \mathcal{B}((x,y) + \mu, c_2) - \operatorname{sgn} \mathcal{B}((x,y) + \mu, c_1)}{2} q^{P((x,y)+\mu)/2}.$$

For the exact comparison with Zwegers' convention, put

$$Q_0(X, Y) := \frac{1}{2}(X^2 - 5Y^2), \quad \mathcal{B}_0((X, Y), (X', Y')) := XX' - 5YY'.$$

Thus $Q_0(v) = \frac{1}{2}v^t A v$ for $A = \operatorname{diag}(1, -5)$, and $\mathcal{B} = 2\mathcal{B}_0$. Set

$$a := \mu = \left(\frac{1}{2}, \frac{1}{10}\right), \quad b := \left(\frac{1}{2}, -\frac{1}{10}\right).$$

For $v = (x, y) + a$ one has

$$(4) \quad e^{2\pi i \mathcal{B}_0(v, b)} = e^{3\pi i/5} (-1)^{x+y}.$$

Theorem 6.4 (Exact Zwegers component). *Let $F(\tau) := q^{1/10} B(q)$. In the notation of Definition 2.1 of [12],*

$$(5) \quad \widehat{F}(\tau) = \frac{1}{2} e^{-3\pi i/5} \vartheta_{a,b}^{c_2, c_1}(\tau).$$

Its holomorphic sign part is F . Moreover

$$(6) \quad \widehat{F}(\tau + 1) = e^{\pi i/5} \widehat{F}(\tau)$$

and

$$(7) \quad \widehat{F}(-1/\tau) = \frac{\tau}{2\sqrt{5}} \sum_{m=0}^4 \vartheta_{b+(0, m/5), -a}^{c_2, c_1}(\tau).$$

Thus $\widehat{F}$ belongs to the finite characteristic orbit supplied by Zwegers' weight-one transformation formulas.

Proof. Equation (3), the identity (4), and the fact that $\mathcal{B}$ and $\mathcal{B}_0$ have the same signs give (5). Both vectors are negative for Q_0 and lie in the same negative component, since

$$\mathcal{B}_0(c_1, c_2) = -15 < 0;$$

Proposition 6.2 shows that no shifted lattice point lies on either boundary. Zwegers' completion theorem applies.

For the T -transformation, Corollary 2.9 of [12] gives the factor $e^{-2\pi i Q_0(a)} = e^{-\pi i/5}$ and replaces b by $a + b + \frac{1}{2}(1, 1)$. The corollary's second exponential factor is 1, because $\mathcal{B}_0(A^{-1}A^*, a) = \mathcal{B}_0((1, 1), a) = 0$. The new characteristic differs from b by $(1, 3/5)$, an element of $A^{-1}\mathbb{Z}^2$, and the characteristic shift contributes $e^{2\pi i \mathcal{B}_0(a, (1, 3/5))} = e^{2\pi i/5}$. Their product is $e^{\pi i/5}$, proving (6).

For S , the same corollary sums over $A^{-1}\mathbb{Z}^2/\mathbb{Z}^2$, represented by $(0, m/5)$ for $0 \leq m < 5$. Here $\det A = -5$ and $\mathcal{B}_0(a, b) = 3/10$, so its prefactor is $(\tau/\sqrt{5})e^{3\pi i/5}$. Multiplication by the constant in (5) gives (7).

There is a further reduction not needed for the transformation formula. For $a'_m = b + (0, m/5)$, the characteristic $a'_3 = (1/2, 1/2)$ is odd relative to $-a$, so $\vartheta_{a'_3, -a}^{c_2, c_1} = 0$.

Corollary 2.9(1)–(3), applied to integer translation, dual translation, and simultaneous negation of the characteristics, also gives

$$\vartheta_{a'_0, -a}^{c_2, c_1} = e^{-\pi i/5} \vartheta_{a'_1, -a}^{c_2, c_1}, \quad \vartheta_{a'_2, -a}^{c_2, c_1} = e^{3\pi i/5} \vartheta_{a'_4, -a}^{c_2, c_1}.$$

Substitution yields the displayed two-component form. $\square$

7. ONE-SIDED COMPLETION AND THE MIXED BOUNDARY DERIVATIVE

We now distinguish the two boundary corrections. The argument is concrete and does not infer an analytic category merely from the inequality $P(c_1) \neq P(c_2)$.

Lemma 7.1 (Odd-theta cancellation). *For every $a > 0$ and every τ in the upper half-plane,*

$$\sum_{n \in \mathbb{Z}} (-1)^n e^{2\pi i \tau a(n+1/2)^2} = 0.$$

Proof. The involution $n \mapsto -1 - n$ preserves $(n + \frac{1}{2})^2$ and changes $(-1)^n$ to its negative. It has no integral fixed point, so the terms cancel in pairs. $\square$

Proposition 7.2 (The c_1 correction vanishes). *In the Zwegers completion of Theorem 6.4, the boundary correction attached to $c_1 = (0, 1)$ is identically zero.*

Proof. The vector $e_1 = (1, 0)$ satisfies

$$\mathcal{B}(c_1, e_1) = 2(0 \cdot 1 - 5 \cdot 1 \cdot 0) = 0, \quad |\det(c_1, e_1)| = 1,$$

so the lattice splits unimodularly in the c_1 and e_1 directions. On each longitudinal fiber the transverse variable is x with shift $\frac{1}{2}$, and the character restricts as $(-1)^{x+y} = (-1)^y \cdot (-1)^x$, i.e. to $(-1)^n$ on the transverse fiber. The correction therefore factors into its longitudinal error-function term times the transverse series in Lemma 7.1. The latter vanishes identically, so the complete c_1 correction vanishes. $\square$

For the second boundary, take $e_2 = (3, -1)$. Then $\mathcal{B}_0(c_2, e_2) = 0$ and $|\det(c_2, e_2)| = 4$, so the boundary lattice has four residue components. For $v = (x, y) + a$, write

$$n := 5x + 15y + 4, \quad T := 3x + 5y + 2.$$

The inverse formulas are

$$x = \frac{3T - n - 2}{4}, \quad y = \frac{3n - 5T - 2}{20},$$

so the integer lattice is the disjoint union, for $j \in \mathbb{Z}/4\mathbb{Z}$, of the pairs satisfying

$$n \equiv 5j + 4 \pmod{20}, \quad T \equiv 3j + 2 \pmod{4}.$$

Moreover

$$(8) \quad v = \frac{n}{20} c_2 + \frac{T}{4} e_2, \quad Q_0(v) = \frac{T^2}{8} - \frac{n^2}{40}, \quad (-1)^{x+y} = (-1)^j.$$

Define the transverse and longitudinal theta series

$$(9) \quad \theta_j(\tau) := \sum_{\substack{T \in \mathbb{Z} \\ T \equiv 3j+2 \pmod{4}}} q^{T^2/8}, \quad g_j(\tau) := \sum_{\substack{n \in \mathbb{Z} \\ n \equiv 5j+4 \pmod{20}}} n q^{n^2/40}.$$

These are vector-valued unary theta components of weights $\frac{1}{2}$ and $\frac{3}{2}$, respectively.

Proposition 7.3 (Exact boundary derivative). *Put $Y = \text{Im}(\tau)$. For the completion (5),*

$$(10) \quad \frac{\partial \widehat{F}}{\partial \bar{\tau}} = -\frac{i}{4\sqrt{10Y}} \sum_{j=0}^3 (-1)^j \theta_j(\tau) \overline{g_j(\tau)}.$$

Consequently

$$(11) \quad \xi_1(\widehat{F}) = -\frac{\sqrt{Y}}{2\sqrt{10}} \sum_{j=0}^3 (-1)^j \overline{\theta_j(\tau)} g_j(\tau).$$

Both expressions are nonzero, and (11) is not holomorphic.

Proof. For the second boundary,

$$\mathcal{B}_0(c_2, v) = -n, \quad Q_0(c_2) = -10.$$

Zwegers' error kernel is therefore $E(-n\sqrt{Y/10})$. Since $E'(z) = 2e^{-\pi z^2}$,

$$\frac{\partial}{\partial \bar{\tau}} E(-n\sqrt{Y/10}) = -\frac{in}{2\sqrt{10Y}} e^{-\pi n^2 Y/10}.$$

Insert this identity in (5), use (8), and group by j . The constant phase in (4) cancels the prefactor in (5), yielding (10). The c_1 derivative contributes zero by Proposition 7.2. Taking complex conjugates and multiplying by $2iY$ gives (11).

For nonvanishing, the $j = 3$ component has the unique lowest antiholomorphic exponent: $n = -1$ contributes $-\bar{q}^{1/40}$ to $\bar{g}_3$, while $T = -1$ contributes $q^{1/8}$ to θ_3 . No other g_j has exponent $1/40$. Hence the coefficient of $q^{1/8}\bar{q}^{1/40}$ in (10) is nonzero. Formula (11) contains the corresponding nonconstant factor $\bar{\theta}_3$, so it is not holomorphic. $\square$

Proof of Theorem 1.1. The transformation statements are Theorem 6.4; the differential statements are Proposition 7.3. For a weight-one harmonic Maass form, the ξ_1 image is holomorphic of weight one. Since (11) is not holomorphic, this completion is not harmonic. Thus this construction does not establish ordinary weight-one mock modularity in the standard harmonic-Maass sense; see [4] for the operator convention and [2] for the harmonic-Maass framework. The calculation alone does not rule out the existence of a different harmonic completion of F . $\square$

8. COMPARISON WITH THE MAASS CASES

The following table separates the proved differential calculation from the structural comparison that motivates it.

Feature	ADH/Cohen over $\mathbb{Q}(\sqrt{6})$	B over $\mathbb{Q}(\sqrt{5})$
Discriminant	24	20
Negative-vector norms	equal, automorph-related	-5 and -20
Boundary corrections	cancel in the Maass construction	one vanishes, one survives
Differential image	Maass-type cancellation	mixed theta expression
Coefficient mechanism	Hecke-character descent	classical descent obstructed
Result	Maass waveform	nonharmonic indefinite completion

In the ADH/Cohen case, the relevant equal-norm vectors are related by a Pell automorphism, and the associated boundary terms combine to give the Maass construction [1, 6, 11]. Here the norm inequality

$$P(c_1) = -5 \neq -20 = P(c_2)$$

prevents any form-preserving automorphism from carrying one vector to the other. This explains why the classical symmetry is unavailable, but it is not used as a converse theorem. The analytic category is decided instead by the exact differential image in Section 7.

The transverse parity gives a second concrete contrast. At c_1 the primitive perpendicular direction is $(1, 0)$, whose coordinate sum is odd, and the shifted transverse theta is killed by Lemma 7.1. At c_2 the perpendicular direction is $(3, -1)$, whose coordinate sum is even, and the four residue components survive. This concrete parity statement is sufficient for the present example; we do not assert a general if-and-only-if criterion for every shifted Pell lattice.

Zwegers' mock Maass theta functions [11] already show that real-quadratic sign kernels can have nonholomorphic completions outside the classical Maass-waveform examples. The contribution here is the exact placement of this cone series in that landscape: the specific shift, the corrected boundary signs, the one-sided vanishing, and the level-40 exponent normalization of the longitudinal theta components. In particular, the paper does not present the general theory of mock Maass theta functions as new.

9. COEFFICIENT DATA, FORMALIZATION SCOPE, AND OPEN QUESTIONS

9.1. A coefficient witness. The arithmetic function obtained from B_N on integers $M \equiv 1 \pmod{10}$ is not multiplicative. The smallest displayed witness from the companion paper is

$$B_1 = 1, \quad B_3 = -2, \quad B_{34} = 3,$$

so that, for $11 \cdot 31 = 341$,

$$B_{(341-1)/10} = 3 \neq -2 = B_{(11-1)/10} B_{(31-1)/10}.$$

This fact is logically separate from Theorem 1.1: the analytic classification uses the exact completion and its derivative, not a contrapositive argument from nonmultiplicativity.

9.2. Computational checks. Direct enumeration of the cone and slab definitions agrees coefficientwise well beyond the range needed for the examples in this paper. The accompanying script `slab.check.py` performs an independent exact-integer check for every $0 \leq N \leq 50$, prints the two coefficients, and reports pass or fail for each N . Its bounds are derived from the cone inequalities rather than chosen as a search radius.

9.3. Coefficient statistics. Through $N \leq 10^5$: $\max_N |B_N| = 7$, first attained at $N = 27252$; among the 19,617 primes $p \equiv 1 \pmod{10}$ with $(p-1)/10 \leq 10^5$, every value $B_{(p-1)/10}$ lies in $\{-2, -1, +1, +2\}$, with magnitude 1 for 13,097 primes (66.8%) and magnitude 2 for 6520 primes (33.2%); there are 4904 indices at which $10N+1$ is a norm but $B_N = 0$, all composite. Empirically $|B_N| = O(d(10N+1))$, a divisor-type bound. The companion paper explains the observed prime-magnitude split using an invariant that combines an archimedean sector label with a congruence label. Its density statement is logically separate from the present analysis.

9.4. Lean scope. The Lean development checks the finite algebraic identities used for the cone coefficients, the involution, the slab decomposition, the Pell-coordinate formulas, the boundary-sign certificates, and a finite residue-support surrogate, with no admitted lemmas in that certificate layer. The coefficient value $B_{34} = 3$ is checked by kernel evaluation and depends only on Lean’s usual three axioms. It also checks the full Laurent-series row-model factorization: the two Jacobi legs, their reindexing, the formal Jacobi triple product, the H -level convolution, even support, and the halved-variable identity in Theorem 5.7. On the analytic side, Lean defines Zwegers’ normalized error kernel, proves its derivative and boundary limits, constructs the convergent general-characteristic indefinite theta and unary theta components, and proves the analytic odd-theta cancellation used for the transverse c_1 factor. It formalizes the complete specialized proof of Zwegers’ Lemma 2.8, including differentiation under the integral, integration by parts, the two Gaussian evaluations and the constant-by-oddness argument. It also proves genuine two-dimensional Poisson summation by periodization on the two-torus, specializes it to the completed kernel, and derives Corollary 2.9(5), (7), and its two-term reduction. The T -law, the exact $\partial_{\bar{\tau}}$ and ξ_1 formulas, and the explicit nonharmonicity witness are likewise kernel-checked. The more interpretive assertion that this explicit sum is the canonical completion of its holomorphic part, and a closed vector-valued representation containing the full characteristic orbit, are not formalized and are not claimed here. The arithmetic foundations and their machine-checked scope are described in the companion paper [9].

Remark 9.1 (Scope of the row-model provenance). The coefficient theorem `mk_factorization`, in namespace `Ch10` within `QseriesFormalization`, proves the finite convolution for `MKcoeff`. It is paired with `coneDiffH.two_mul`, which identifies the even cone coefficients with B_N , and `coneDiffH.odd`, which proves odd-level vanishing. The Laurent-series assembly is `paper2MKLaurent_eq_jacobi_sq_mul_cone`; the injective even-exponent compression concludes with `paper2RowLaurent_eq_jacobi_sq_mul_B`, which is exactly the boxed formula in Theorem 5.7. No source-faithfulness theorem in the repository currently identifies `MKcoeff` with the older Chapter-10 object

`rhoEvenSourceOldRow_missing`.

Accordingly, no theorem about the full function Θ_{10} is deduced here.

9.5. A family suggested by the example. For squarefree $d \equiv 5 \pmod{8}$, the prime 2 is inert in $\mathbb{Q}(\sqrt{d})$. Elementary mod-8 arithmetic on the Pell equations $b^2 - da'^2 = 1$ and $b^2 - da'^2 = 4$ then forces alternating and constant transverse characters, respectively, at the two boundary directions, which suggests the one-sided pattern of Section 7; this parity computation has been checked numerically for $d = 5, 13, 29, 37, 53, 61$. The computation alone does not identify the relevant boundary vectors for every d , so we state only a conjecture.

Conjecture 9.2. *Let $d \equiv 5 \pmod{8}$ be squarefree, and suppose a conductor-2 dissection over $\mathbb{Q}(\sqrt{d})$ produces a signature- $(1, 1)$ shifted lattice with boundary vectors $c_1(d)$, $c_2(d)$ of norms $-d$ and $-4d$, positioned as in the $d = 5$ model of Section 6 (in particular $\mathcal{B}(c_1(d), c_2(d)) < 0$ and the shifted lattice avoids the boundary lines). Then the correction at $c_1(d)$ vanishes by the odd pairing of Lemma 7.1, the correction at $c_2(d)$ survives, and the residual indefinite theta function has a nonzero mixed boundary derivative.*

9.6. Open questions.

- (1) What is the minimal closed vector of characteristics containing $\widehat{F}$ under S and T , and what is its smallest congruence subgroup realization?
- (2) Which other classical dissections produce one-sided or two-sided boundary corrections?
- (3) Are the coefficients B_N unbounded, and can one prove a divisor-type upper bound from the norm interpretation?
- (4) For which squarefree d can the conductor-2 boundary vectors be determined uniformly enough to decide Conjecture 9.2?

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